

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9	(a)	(i)		Aberystwyth has less radon than Penzance but more than Chester (1) OWTTE and the pie chart for location A has less radon than location C but more than location B (1) [so Adam is correct]			2	2		
		(ii)	I	$\frac{30}{60} = 0.5$ [cps]		1		1	1	1
			II	15		1		1	1	1
		(iii)		Man-made sources are only 14.3% or 14.5% (1) this is lower than natural sources (1) ORA so disagree Conclusion must be present to award both marks OR Radon [is a natural source and it] is 50% (1) So man-made sources can't be more than 50% (1) so disagree Conclusion must be present to award both marks OR Correctly named sources of natural radiation are stated with percentage stated (1) So there is a larger proportion from natural sources (1) so disagree Conclusion must be present to award both marks			2	2		
	(b)	(i)		Attempt at adding her total dose (1) Total dose is $2.7 + 0.18 + 2 \times 0.09 + 0.005 + 6.6 = 9.665$ [units] [this is less than 20] so it is still safe [for her to work with radiation] (1)			2	2	1	

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		(ii)		{16 hours of flying / 2 flights} (1) gives the same [exposure] as a power station worker [gets in one year] (1) [so disagree] OR 3 or more flights or more than 24 hours of flying (1) gives more [exposure] than a power station worker [gets in one year] (1) [so disagree] Accept: In 1 year airline pilots receive 32.85 [units] (1) gives more [exposure] than a power station worker [gets in one year] (1) [so disagree]			2	2		
				Question 9 total	0	2	8	10	3	2